Graphical Representation

Histograms, ogives, polygons, stem-and-leaf plots, bar and pie charts, and boxplots.

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1 Histogram

Adjoining rectangles for a **continuous** frequency distribution; class boundaries on the x-axis.

Equal widths	Unequal widths
Bar height = **frequency**	Bar height = **frequency density**

Frequency density

$$\text{FD} = \frac{\text{Class frequency}}{\text{Class width}}$$

The mode can be estimated graphically from the histogram.

2 Ogives (cumulative frequency curves)

Type	Plot	Shape
Less-than	LCF against **upper** class boundaries	Rises
Greater-than	GCF against **lower** class boundaries	Falls

The two ogives **intersect at the median**. The median can also be read from the less-than ogive at $N/2$.

3 Frequency polygon

Plot frequencies at **class midpoints** and join them with straight lines. Useful for comparing two distributions on one graph.

4 Stem-and-leaf plot

Each value split into a **stem** (leading digits) and a **leaf** (trailing digit). Shows the shape **and keeps every observation**.

e.g. 107 → stem 10, leaf 7.

5 Bar charts

- **Simple**: one value per category
- **Component** (sub-divided): parts of each category
- **Grouped** (multiple): several values within each category

For categorical data, so bars are separated by gaps.

6 Histogram vs bar chart

	Histogram	Bar chart
Data	Continuous quantitative	Categorical / discrete
Bars	Adjacent, no gaps	Separated by gaps
Area	Represents frequency	Only height matters
Order	Fixed by the scale	Categories can be reordered

7 Pie chart

A circle split into sectors showing **parts of a whole**. The full circle is **360°**.

Sector angle

$$\frac{\text{Value of category}}{\text{Sum of all values}} \times 360^\circ$$

Best with a few categories that add up to a meaningful total.

8 Boxplot (box-and-whisker)

Shows the **five-number summary**: minimum, Q_1, **median** (line inside the box), Q_3, maximum, plus the spread and potential outliers.

Interquartile range

$$IQR = Q_3 - Q_1$$

Lower fence

$$Q_1 - 1.5 \times IQR$$

Upper fence

$$Q_3 + 1.5 \times IQR$$

Points beyond the fences are flagged as **potential outliers**.

! Common mistakes

- × Unequal class widths with raw frequencies as heights.
- × Plotting the less-than ogive at **lower** boundaries.
- × Leaving gaps between histogram bars.
- × Pie charts for data that are not parts of a whole.

? Fill in the blanks

1. Classes 0–10 and 10–30 each have frequency 20; their histogram bar heights should be _____ and _____. **[2 and 1 (frequency densities)]**
2. A crop occupying 25% of the cropped area gets a pie-chart sector of _____°. **[90]**
3. To read the median from a less-than ogive with N = 60, draw a horizontal line from cumulative frequency _____. **[30]**
4. If Q_1 = 20 and Q_3 = 30, any observation above _____ is flagged as a potential outlier. **[45]**
5. Area under three crops in each district, stacked inside one bar per district, is shown by a _____ bar chart. **[component]**
6. A display that shows the shape of the data yet lets you recover every observation is the _____ plot. **[stem-and-leaf]**

Source: *Textbook of Agricultural Statistics*, Ch. 3 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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